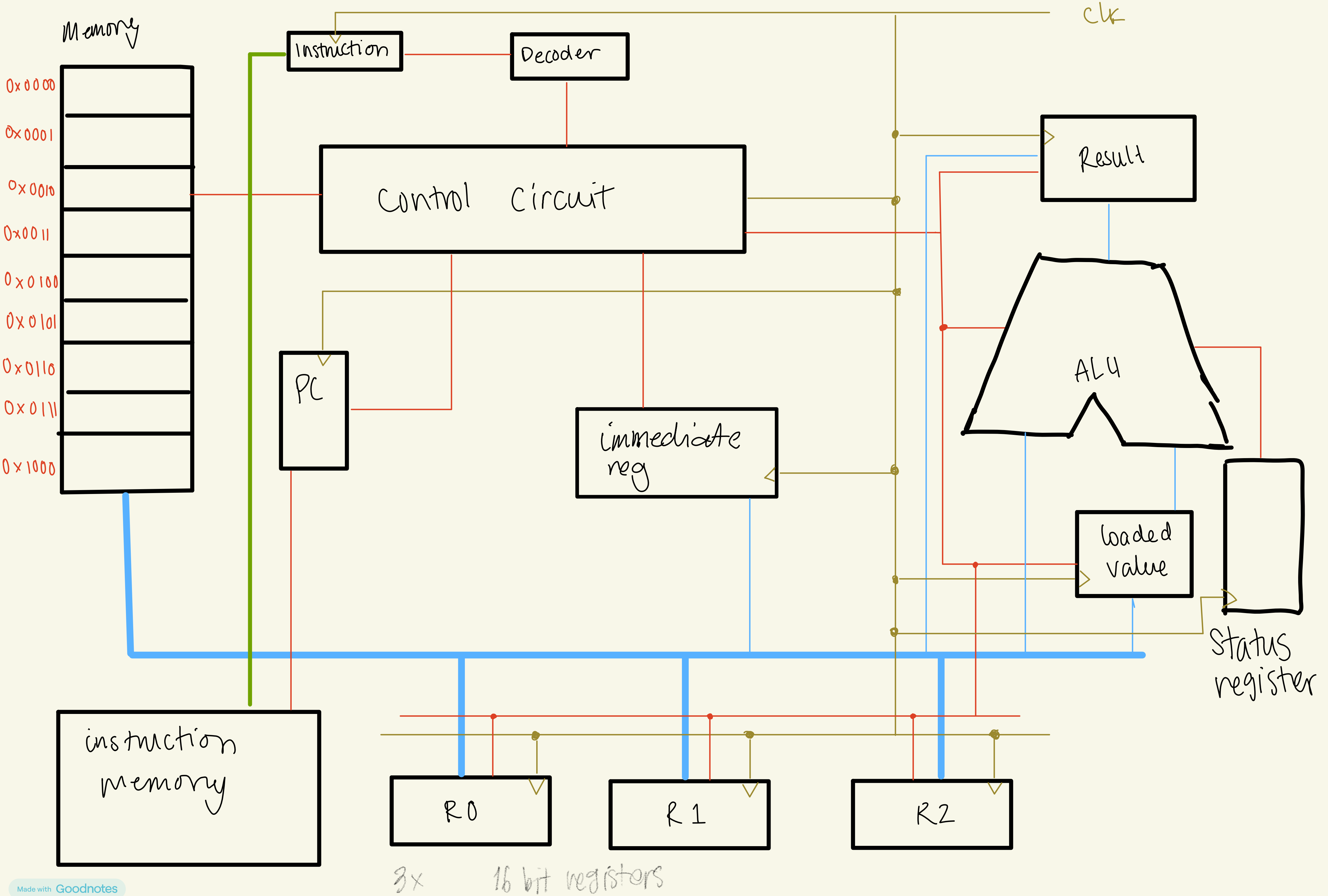
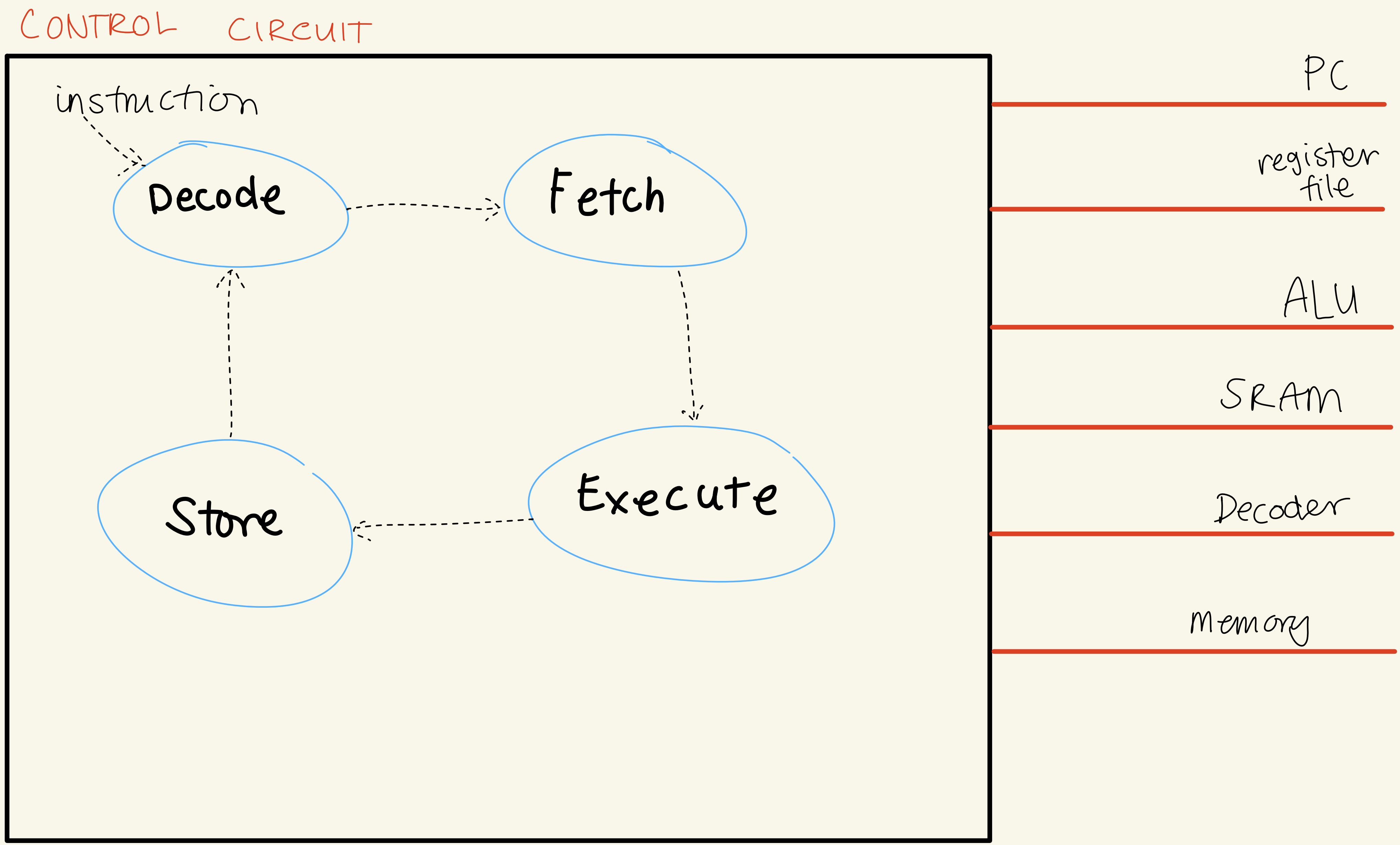


Memory
0x0000
0x0001
0x0010
0x0011
0x0100
0x0101
0x0110
0x0111
0x1000
Instruction
Decoder
clk
Control Circuit
Result
PC
ALU
immediate reg
loaded value
Status register
instruction memory
R0
R1
R2
3x 16 bit registers

Opcode

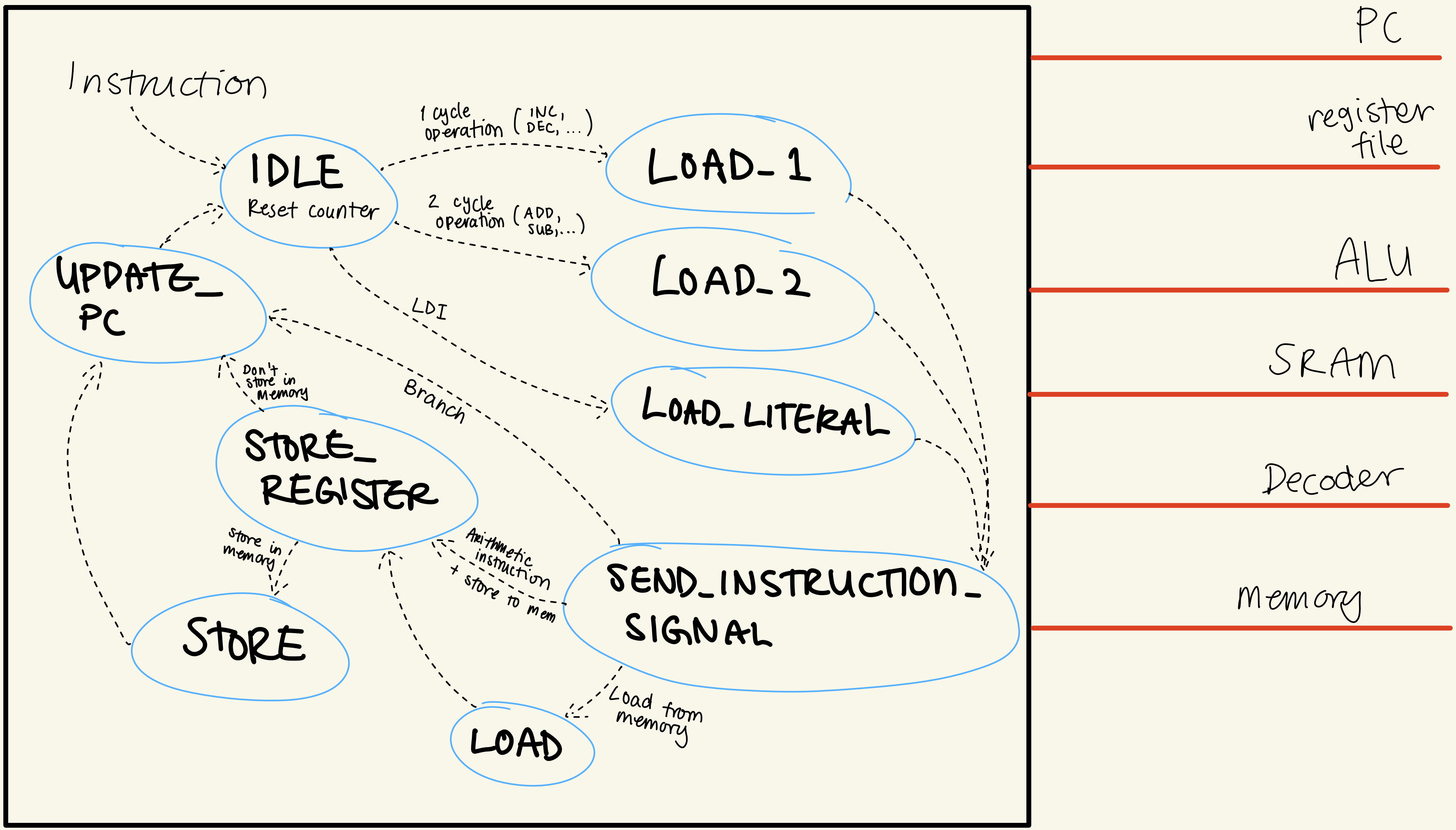
INSTRUCTION	PREFIX	SUFFIY	ARGUMENTS
Add	0xA	0x1	R[n:0] x2
Sub	0xA	0x2	R[n:0] x2
Inc	0xA	0x3	R[n:0]
Dec	0xA	0x4	R[n:0]
Lsr	0xA	0x5	R[n:0] & immediate
Lsl	0xA	0x6	R[n:0] & immediate
Xor	0xA	0x7	R[n:0] x2
And	0xA	0x8	R[n:0] x2
Nor	0xA	0x9	R[n:0] x2
Not	0xA	0x10	R[n:0]
Or	0xA	0x11	R[n:0] x2
Ldi	0xD	0x1	R[n:0] & immediate
Ld	0xD	0x2	R[n:0] & mem addr
St	0xD	0x3	R[n:0] & mem addr
Mov	0xD	0x4	R[n:0] x2
Jmp	0xC	0x1	PMEM & SREG
Je	0xC	0x2	PMEM & SREG
Jg	0xC	0x3	PMEM & SREG
JL	0xC	0x4	PMEM & SREG
CP	0xC	0x5	R[n:0] x2

High-level FSM

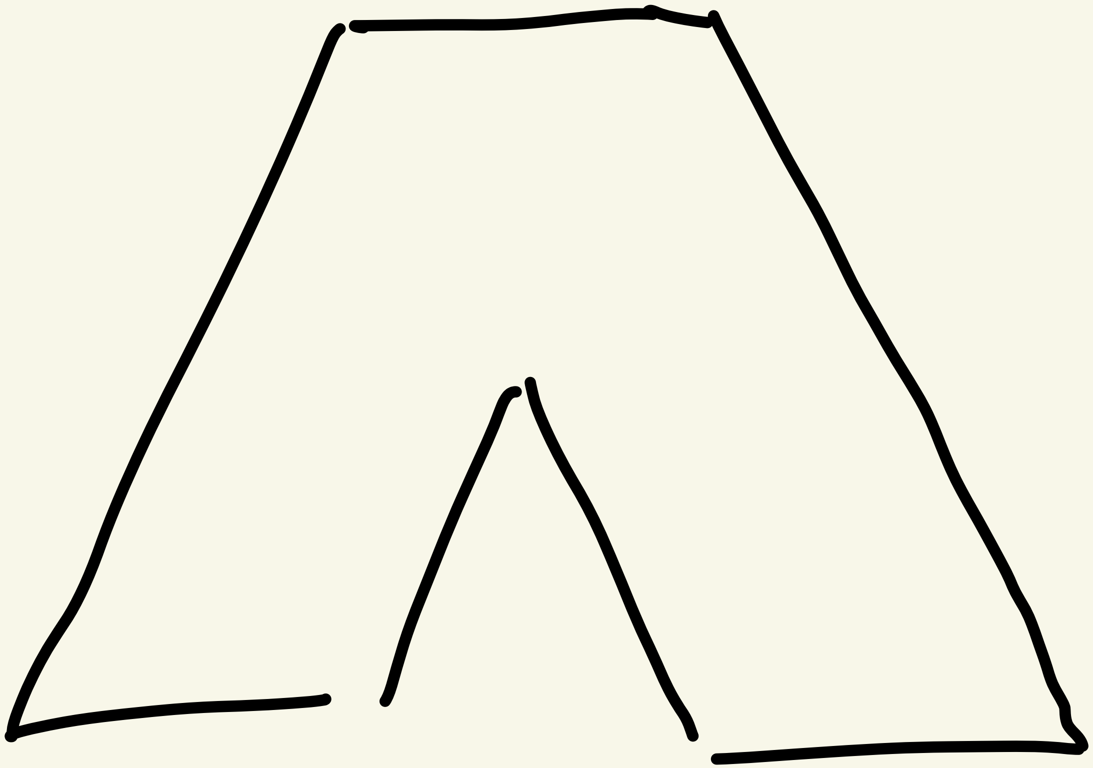


Low-Level FSM

CONTROL CIRCUIT



ALU



Supported instructions

- Adder
  - ↳ add
  - ↳ 2's complement ← 2 clock cycles
    1. flip bits
    2. add 1
- Shift register (16 D flip-flops)
  - ↳ LSL
  - ↳ LSR
  } ask Rudy!
- Bitwise operators
  - ↳ OR, AND, XOR, NOT
    - ↳ simply a truth table
- Comparison
  - ↳ Status register

again ask Rudy